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Export: Sensor Data

This help center article explains how to export 'sensor data' (i.e., the raw data and/or metric data from eye tracking, facial expression analysis, EDA, EEG, etc).

What exactly is the 'sensor data' export?

The sensor data export allows users to export the *unprocessed* and/or *processed* sensor data from eye tracking, facial expression analysis, EDA, EEG, etc. in .csv format. This could be (1) the 'raw' sensor data (if provided by the hardware systems used), (2) the 'processed' sensor data provided by the hardware systems (if provided by the hardware systems used), and/or (3) the 'processed' sensor data provided by iMotions signal processing algorithms (if R-Notebooks were applied). In addition, the sensor data export files may include various types of recording 'events' (e.g., stimulus onsets/offsets, live markers, mouse clicks, annotations, etc.).

This type of export file is mostly useful if you wish to apply your signal processing algorithms in external software (e.g., specialized EEG software).

Below, we explain the format of the sensor data export files in detail.

In brief, the sensor data export files represent the *continuous* data (i.e., timeline data) of *individual* respondents (i.e., one file per respondent), and always include:

- A timeline (i.e., the iMotions synchronization timeline)
- Slideshow events
- Sensor data (i.e., *at least* one data column of sensor data obtained from *at least* one hardware system)

In addition to that, the sensor data export files may also include:

- Recording events (e.g., live markers, markers received through the API, mouse clicks, scroll events)
- Post-imported data (e.g., R-processed data)
- Annotations
- Scenes
- Gaze-mappings

It is worth noting, thereby, that the sensor data export files orient the sensor data (i.e., from one or multiple devices) on a common timeline. In other words: the sensor data is 'synchronized'. By synchronizing the sensor data, the iMotions software tries to maximally support cross-modal types of analyses, including advanced types of analyses that take place in external software (e.g., specialized EEG software). However, apart from synchronizing the sensor data, the export files also maximally respect the original sensor data as provided directly by the hardware systems themselves (i.e., no down-sampling, filtering, smoothing, etc).

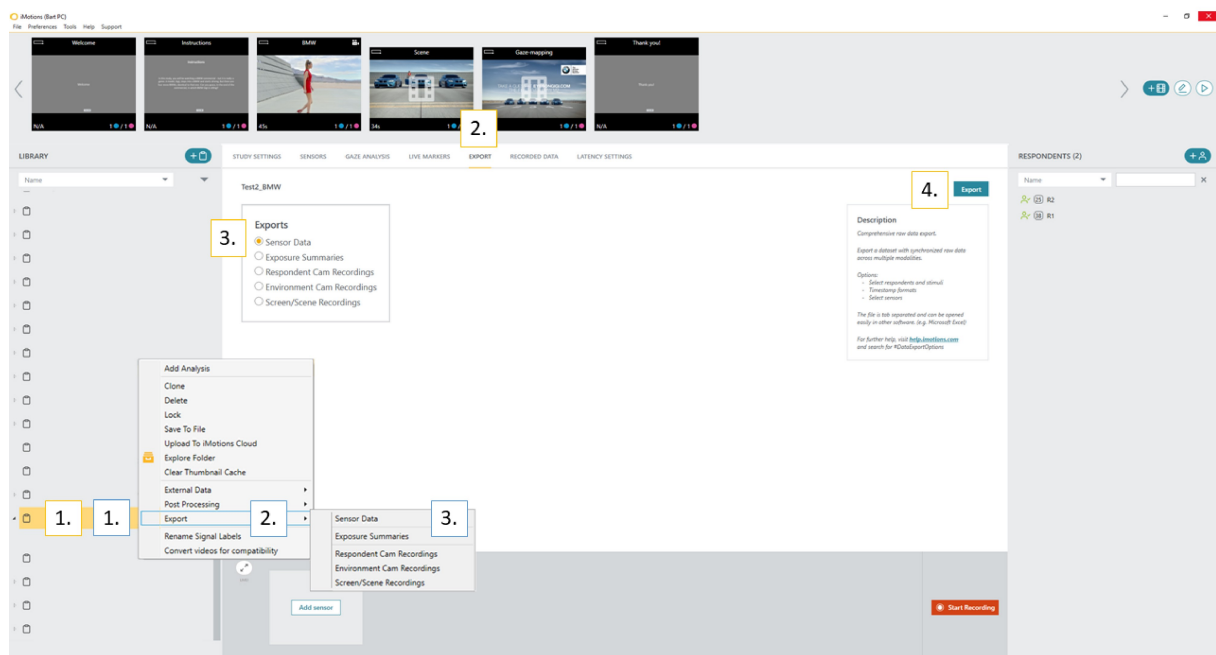
Click [here](#) for more information about the synchronization and timestamping method used by the iMotions software.

How to:

Exporting the sensor data is done using the sensor data export editor - a functionality that allows users to customize and preview the export files according to their specific needs. This export editor can already be accessed *immediately after a recording has ended*, or later, *from within an existing analysis*.

Below we first explain how to access the export editor, and the basic steps involved in exporting the sensor data from the export editor.

How to open the sensor data export editor immediately after a recording has ended



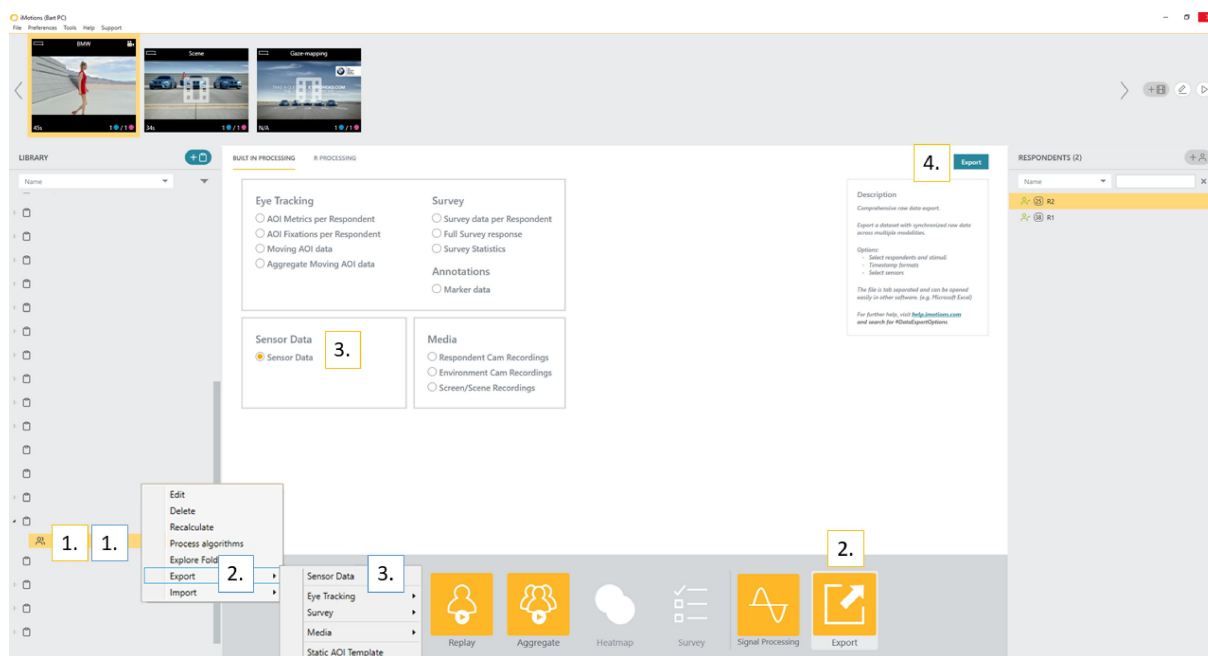
How to open the sensor data export editor:

1. Click the name of the study in the Library
2. Navigate to the tab 'Export'
3. Select the option 'Sensor data'
4. Click the blue 'Export' button in the top-right corner.

Alternatively:

1. Right-click the name of the study in the Library
2. In the right-click menu, choose 'Export'
3. Then, click 'Sensor data'

How to open the sensor data export editor from within an existing analysis



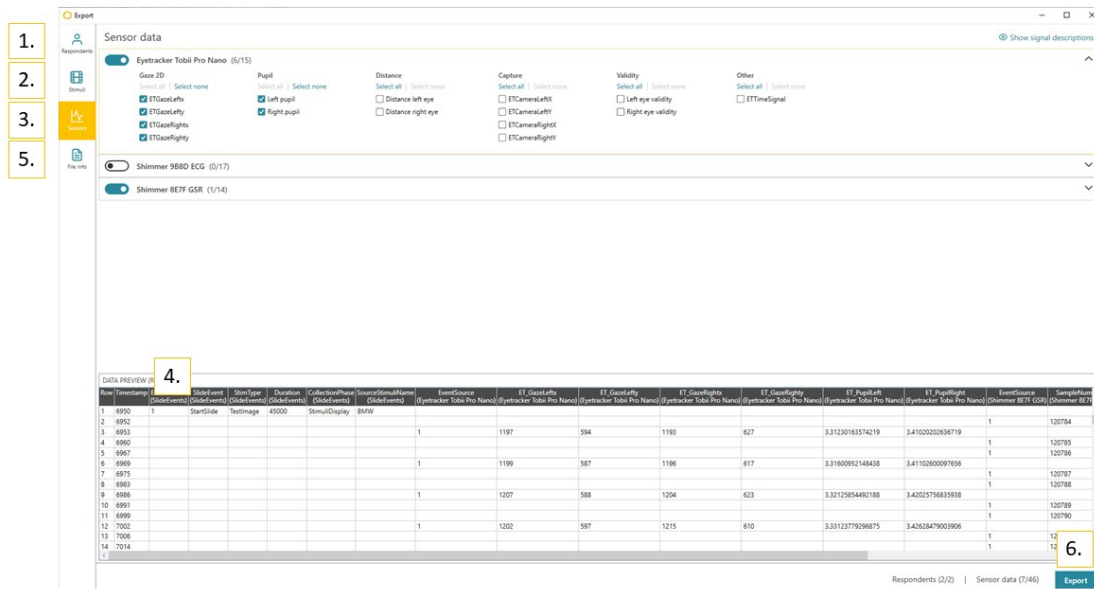
How to open the sensor data export editor within an analysis:

1. Click the name of the analysis in the Library
2. Click the 'Export' button in the analysis menu
3. Select the option 'Sensor data'
4. Click the blue 'Export' button in the top right corner

Alternatively:

1. Right-click the name of the *analysis* in the Library
2. In the right-click menu, choose 'Export'.
3. Then, click 'Sensor data'

How to use the sensor data export editor (basic steps)



Once the sensor data export editor is opened, it takes a few more steps to export the sensor data export-files using the export editor:

1. On the 'Respondent' page, select the relevant respondents
2. At the 'Stimuli' page, select the relevant stimuli
3. On the 'Sensors' page, select the relevant data columns. (Note that the processed sensor data produced by R-Notebooks or other algorithms in iMotions are not listed here, but exported from the 'File info' page).
4. Preview the sensor data export-file
5. At the 'File info' page, further customize the format of the export file by ticking the relevant options.
6. Choose the desired folder for saving, and click the blue 'Export' button in the bottom-right corner

The format (color-coded)

Below, we further explain the available options for customizing the sensor data export files. Here, we first explain the format of the sensor data export files in further detail.

Concerning the format of the export files, it is worth noting that the sensor data export produces *individual* export files (i.e., one file per respondent), which contain the *continuous* data (i.e., timeline data) recorded during the experiment – entirely consistent with how the data was originally recorded. That means that the sensor data export files are typically very long files (i.e. many rows of data), because every row of data in the export file refers to a unique moment on the timeline. Specifically, in the case of a recording device (e.g., an eye-tracker) with a sampling rate of 120 Hz, then every second of the recording 120 rows of data are added to the recording, and hence, to the sensor data export file.

This principle summarizes the basic format of the data (#DATA).

In addition to #DATA, the export file may also include the sections #INFO and #METADATA; the sections are color-coded in the illustration below.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	#INFO																					
2	#Study name																					
3	#Respondent Name																					
4	#Respondent Age																					
5	#Respondent Gender																					
6	#Respondent Group																					
7	#Motions version																					
8	#Current culture																					
9	#Recording time																					
10	#Export time																					
11	#Sensor info																					
12	#Sensor info																					
13	#Sensor info																					
14	#Sensor info																					
15	#																					
16	#METADATA																					
17	#Device																					
18	#Category																					
19	#Description																					
20	#Unit																					
21	#Data format																					
22	#Instance																					
23	#Short unit																					
24	#Group																					
25	#Provided By																					
26	#Channel identifier																					
27	#																					
28	#DATA																					
29	Row	Timestamp	EventSource	SlideEvent	StimType	Duration	CollectionPhase	SourceStimulNam	EventSource	< Data >	< Data >	...	EventSource	< Data >	< Data >	...	Annotation1	Annotation2	...	ImageScene	VideoScene	...
30	1	3444		1 StartLife	Testimage	3000000	StimulDisplay	Welcome														
31	2	3446																				
32	3	3448																				
33	4	3450																				
34	5	3452																				
35	6	3454																				
36	7	3455																				
37	8	3456																				
38	9	3457																				
39	10	3459																				
40	12	3461																				
41	13	3463																				
42	14	3465																				
43	15	3467																				
44	16	3469																				
45	17	3471																				
46	18	3473																				
47	19	3475																				
48	20	3477																				
49	21	3479																				
50	22	3481																				
51	23	3483																				

Figure. Color-coded illustration of the sensor data export files.

INFO (green)

Section #INFO provides general information about the recorded data (e.g. name of the study, name of the respondent, recording time, information about the hardware systems used, etc.)

METADATA (yellow)

Section #METADATA provides, for each of the data columns in the export file, explanatory information about the data contained in the respective data columns (e.g., where the data comes from, the category of data, a short description of the data, the unit, etc.)

DATA (blue, red, grey, orange)

Section #DATA, finally, is the essential part of the export file; it contains the sensor data, but also other types of data (e.g., 'events').

> Timeline (blue)

The iMotions 'synchronization' timeline is represented by two data columns in the export file.

Row	Simple row index
Timestamp	Elapsed time since the recording started (in milliseconds)

> Slideshow events (red)

The slideshow events (i.e., the start/end of slides in the iMotions study) are represented by five data columns in the export file. These data columns are preceded by an 'Eventsource' data column, which marks the rows of data representing slide events (TIP: use the 'Eventsource' column for filtering the slideshow events in Excel).

Event source	Boolean which marks slide events (1 = slide event)
--------------	--

SlideEvent*	Four types of events (StartSlide, StartMedia, EndMedia, EndSlide)
StimType	<i>(unimportant)</i>
Duration	Planned duration of the slides (in milliseconds)
collection phase	<i>(unimportant)</i>
SourceStimuliName	User-defined name of the slide

* *StartSlide* = start command is forwarded by iMotions to the media player;
StartMedia = confirmation from the media player that the slide has started;
EndMedia = confirmation from the media player that the slide has ended;
EndSlide = end command is executed by iMotions.

> Sensor data (grey)

The sensor data are represented by a variable number of data columns in the export file. Similarly, as for slideshow events, these data columns are preceded by an 'Eventsource' data column, which marks the rows of data representing data samples from a given device (TIP: use the 'Eventsource' column for filtering the data of a given device in Excel).

Event source	Boolean which marks data samples recorded from a given device (1 = data sample)
--------------	---

<Data column 1>	Sensor data
<Data column 2>	Sensor data
...	...

Note that post-imported data, unknown data received by iMotions over the API, or R-processed 'continuous' data (e.g., filtered data) are represented in the same way as the sensor data provided directly by the hardware systems.

> AOIs, Annotations, Scenes, and Gaze mappings (orange)

If AOIs and/or annotations were placed in the iMotions software, scenes were created, or gaze-mapping algorithms were executed, these AOI, annotations, scene and gaze-mapping intervals can be exported as part of the sensor data export-file.

For every AOI, you receive information on when your AOI was active. This reflects the timeline during which you set the AOI:

- For static stimuli (Images, Survey Slides, Image Scenes, Gaze Mappings, Web Scenes) the AOI is automatically active during the entire stimulus.
- For dynamic stimuli (Videos, Web Recordings, Screen Recordings), you choose yourself during your analysis when the AOI shall be active.

In addition to the information on when your AOI was active, you also receive information about when the respondent actually looked ("gazed") at the AOI. Thus, each AOI is represented with a pair of data columns.

If you want to know when your respondents fixated at the AOI, you can take the additional information from the "Eye Tracker Metrics" section into account, which marks up when a fixation or saccade happened.

<AOI 1> (Active)	Boolean which marks gaze data samples that coincide with the AOI being active regardless of whether the gaze falls within that AOI (1 = AOI is active at this time)
<AOI 1> (Gazed)	Boolean which marks gaze data samples that fall within the AOI (1 = gaze point falls within the AOI)
...	...

Every annotation, scene, or gaze-mapping interval is represented by a unique data column in the data file - with the name of the annotation, scene or gaze-mapping in the header. These data columns contain Boolean values which mark all data samples that fall within the annotation, scene or gaze-mapping interval (TIP: use these data columns for filtering the data falling within specific annotation, scene or gaze-mapping intervals in Excel).

<Annotation 1>	Boolean which marks data samples that fall within the annotation interval (1= data sample during annotation)
<Annotation 2>	Boolean which marks data samples that fall within the annotation interval (1= data sample during annotation)
...	...
<Scene 1>	Boolean which marks data samples that fall within the scene interval (1= data sample during annotation)
...	...

What are the options for customizing the sensor data export files?

Finally, we discuss the options for customizing the sensor data export files in further detail; these options are made available at the various pages of the export editor. Note further that the results can be previewed in the bottom of the export editor before clicking the 'Export' button.

- [Respondents page](#)

Options: It is possible to select *individual* respondents or immediately *all* respondents; for the latter, click 'select all'. Moreover, use the 'sort by' function to easily select a specific *group* of respondents (First sort by a group variable, then click 'select all').

- [Stimuli page](#)

Options: As for the respondents, it is possible to select *individual* stimuli or immediately *all* stimuli; for the latter, click 'select all'. Here, the 'sort by' function may be used to easily select stimuli of a given type (First sort by type, then click 'select all').

- [Sensors page](#)

Options: Concerning the sensor data, it is possible to select *individual* data channels (tick the name of a data channel), the data channels belonging to a given *category* of data (click 'select all' below the category name), or immediately *all* data channels of a given device (tick the button next to the device name).

Also helpful, is the option to 'show signal descriptions' in the top-right corner; if enabled, brief descriptions of the data channels are shown below the names of the data channels.

Preview: As soon as at least one data channel is selected, a preview of the export files is shown at the bottom of the screen.

- [File info page](#)

Options: The 'File info' page hosts a few more options for customizing the format of the sensor data export files in the form of tick-boxes. There are several categories of file information. You can quickly view descriptions of these options by clicking "Show descriptions" in the top right corner.

File Info Show descriptions

File output location: C:\Users\jesa\Documents\iMotions\Export\Mobile\Magazine Shopping_meET_GSR_Merged\Analyses\Analysis\Sensor Data Browse

Study properties & meta data
☒ Study properties
☒ Channel headers
☒ Channel meta data

Additional data
☐ User generated events
☐ Processed and imported data

AOIs
☐ Select none
☒ Include AOI markers
☐ Show separate AOI columns per Stimulus
☐ Combine AOI markers
☐ Show AOI markers as text
☐ Include AOI comments & instances

Annotations
☐ Select none
☒ Include Annotation markers
☐ Show separate Annotation columns per Stimulus
☐ Combine Annotation markers
☐ Show Annotation markers as text
☐ Include Annotation comments & instances

Scene & Gazemappings
☐ Select all
☐ Include Scene & Gaze Mapping markers
☐ Combine Scene & Gaze Mapping markers
☐ Show Scene & Gaze Mapping markers as text
☐ Include Scene & Gaze Mapping instances

Stimulus & event source markers
☐ Select all
☒ Repeat stimulus name
☐ Combine Event Source markers
☐ Show Event Source markers as text

Other options
☐ Select all
☐ Select none
☐ Round timestamp to milliseconds
☐ Merge respondents to one file
☒ Continuously update data preview

DATA PREVIEW (Respondent 2)

Row	Timestamp	EventSource (SlideEvent)	SlideEvent (SlideEvent)	SlideType (SlideEvent)	Duration (SlideEvent)	CollectionPhase (SlideEvent)	SourceStimulusName (SlideEvent)	EventSource (Eyetracker Pupil Invisible Glasses)	ET_TimeSignal (Eyetracker Pupil Invisible Glasses)	ET_PupilLeft (Eyetracker Pupil Invisible Glasses)	ET_PupilRight (Eyetracker Pupil Invisible Glasses)	ET_DistanceLeft (Eyetracker Pupil Invisible Glasses)	ET_DistanceRight (Eyetracker Pupil Invisible Glasses)	ET_Gaze (Eyetracker Pupil Invisible Glasses)
1	321.3601	1	StartSlide	Testimage	12232	StimulusDisplay	Pupil Invisible Glasses							
2	321.500286915804						Pupil Invisible Glasses 1	2169.3601	0	0	-1	-1	596	
3	327.686						Pupil Invisible Glasses							
4	335.498						Pupil Invisible Glasses							
5	341.500286915804						Pupil Invisible Glasses 1	2189.3601	0	0	-1	-1	587	
6	343.311						Pupil Invisible Glasses							
7	351.1235						Pupil Invisible Glasses							
8	358.896						Pupil Invisible Glasses							
9	361.500286915804						Pupil Invisible Glasses 1	2209.3601	0	0	-1	-1	584	
10	366.7485						Pupil Invisible Glasses							
11	374.561						Pupil Invisible Glasses							
12	381.500286915804						Pupil Invisible Glasses 1	2229.3601	0	0	-1	-1	592	
13	382.3725						Pupil Invisible Glasses							

Respondents (2/2) | Sensor data (37/66) Export

Study properties & metadata:

- > Study properties: Tick to export the section #INFO
- > Channel headers: Tick to export a row of headers (#DATA)
- > Channel metadata: Tick to export the section #METADATA

Additional data:

- > User-generated events: Tick to export live markers, keyboard, mouse, and website events. Click [here](#) for more information about live markers
- > Processed and imported data: Tick to export post-imported data and/or the processed sensor data from R-Notebooks

Click [here](#) for more information about the post-import of data

Click [here](#) for more information about R-Notebooks

AOIs: Click [here](#) for more information about AOIs

- > Include AOI markers: Tick to add columns indicating the timeline and dwells of all AOIs
- > Show separate AOI columns per stimulus: Tick to split the AOI column into one new column per stimulus

- > Combine AOI markers: Tick to combine AOI markers into 2 columns corresponding to when the AOI is active and when the AOI is gazed on
- > Show AOI markers as text: Tick to use the name of the AOI rather than the Boolean value '1' to mark when samples occur within an AOI
- > Include AOI comments and instances: Tick to add comments and counters per instance of the AOI

Annotations: Click [here](#) for more information about annotations

- > Include annotation markers: Tick to add columns indicating the timeline and associated data of all annotations
- > Show separate annotation columns per stimulus: Tick to split the annotation column into one new column per stimulus
- > Combine annotations markers: Tick to combine annotation markers into 1 column
- > Show annotation markers as text: Tick to use the name of the annotation rather than the Boolean value '1' to mark when an annotation is active
- > Include annotation comments and instances: Tick to add comments and counters per instance of the annotation

Scene & Gaze mappings: Tick to export scene and gaze-mapping intervals. Click [here](#) for more information about the scenes. Click [here](#) for more information about gaze mapping.

- > Include Scene & Gaze mapping markers: Tick to add columns indicating the timeline and associated data of selected scenes
- > Combine Scene & Gaze mapping markers: Tick to combine the markers for scenes, web scenes, and gaze mapping into 3 columns instead of displaying 1 column per marker
- > Show Scene & Gaze mapping markers as text: Tick to use the name of the scene/gaze mapping marker rather than the Boolean value '1' to mark when these are active
- > Include Scene & Gaze mapping instances: Tick to add a column with a counter per instance of the scene or gaze map

Stimulus & event source markers:

- > Repeat stimulus name: Tick to repeat the name of the slide in the data column 'SourceStimuliName' (#DATA) for every row of data
- > Combine event source markers: Tick to combine markers indicating which devices the current data row originates from into one column
- > Show event source name: Tick to use the name of the device rather than the Boolean value '1' to mark the samples in the data columns 'Eventsource' (#DATA)

Other options:

- > Round timestamp to milliseconds: Tick to round the values in the data column 'Timestamp' (#DATA) to the nearest integer
- > Merge respondents to one file: Tick to export all respondent data in a single file instead of having a separate export file for each respondent
- > Continuously update data preview: Untick to disable the export preview box at the bottom of the file info page

